Patient medical data
↓
Load dataset
↓
Created df
↓
separated X & y
↓
train_test_split
↓
model – Logi R.
↓
train model (fit.)
→ predict probabi.
↓
apply d.b.
↓
predict final class
↓
evaluated using accuracy.
confusion matrix, classification report.

Type of problem

- Supervised
  - Numeric → Regression
  - Categorical → classification
- Unsupervised
  - Clusters (group) → Clustering
  - Dimension (Reduce) → PCA / t sne

predicting continuous numerical value

which category

① Gmail filtering
- Spam
- Not spam

② Hospital diagnosis

classification → predicting categories or class

no. of hours	result
1	fail
2	fail
3	fail
4	pass
5	pass
6	pass

fail → 0
pass → 1

$\boxed{0.6}$

0.5 , 0.4

– 0.4

Linear R.	classifi.
$-\infty$ to $+\infty$	0 and 1
predict numbers	predicts probabilities.
straight line	probability curve.

Logistic Regression.

$y = b_0 + b_1 x$

$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + --- b_n x_n$

$z = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + --- + b_n x_n$

Suppose:- age = 30, salary = 10 lakh, credit score = 750, loan amount = 10 lakh

$z = 5$, $z = -3.5$

Sigmoid function.

Input features → Linear Eqⁿ (z) → Sigmoid function → probability → D.B (0.5)
↓
final class

σ

$\sigma(z) = \dfrac{1}{1 + e^{-z}}$

$\boxed{0 , 1}$

Decision Boundary.

$\boxed{0.5}$

0 - - - - - - . 0.5 - - - - - . 1
negative class — positive class

100 → 35 less H,
↓ pass — fail

$\boxed{0.49}$

$\boxed{0.3}$

$\boxed{CBC}$ → Sugar → $\boxed{150}$, $\boxed{200}$

0 → malignant
1 → benign

enumerate()
↳ index position &

```
fruits = ["Apple", Ba "Mango"]
for i, fruit in enumerate(fruit):
    print(i, fruit)
```

0 Apple
1 Banana
2 Mango

[

["Dog", "Cat", "Lion"]

0 Dog
1 Cat
2 Lion.

1 → Mali
0 → Benign

M → 0
F → 1

M → 1
F → 0

Age	Gender	Disease (Disease)
25	M	0
35	F.	1